

BANJUL YUNDUM, Gambia

WMO: 617010

Lat: 13.338N

Lon: 16.652W

Elev: 95

StdP: 14.65

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.7	63.0	31.8	26.2	80.8	34.7	29.6	80.4	22.4	78.0	20.0	77.7	5.6	90	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	15.3	100.1	68.5	97.0	68.7	95.0	69.6	81.9	88.7	81.1	87.9	80.5	87.1	9.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.9	86.4	79.2	151.8	85.1	78.9	150.3	84.8	45.7	89.5	44.9	88.6	44.3	87.5	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.8	14.8	DB	56.0	105.8	4.9	2.5	52.4	107.6	49.6	109.0	46.8	110.5	43.2	112.3
			WB	48.6	85.6	2.5	1.9	46.8	86.9	45.3	88.0	43.9	89.1	42.1	90.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	77.0	79.2	80.2	79.9	80.5	82.9	81.9	81.1	81.5	83.1	81.3	78.3
	DBStd	3.31	3.07	3.54	4.09	3.71	3.19	2.17	2.02	1.87	1.92	1.94	2.40	2.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11159	838	817	936	897	946	987	988	964	945	1025	939	877
	CDD65	5684	373	397	471	447	481	537	523	499	495	560	489	412
	CDH74	52613	3194	3585	4054	3499	3889	5484	5320	4661	4762	5994	4640	3531
	CDH80	22621	1612	1976	2249	1732	1712	2169	1763	1413	1644	2431	2167	1753
(14) Wind	WSAvg	6.7	7.7	8.4	8.7	8.9	8.4	7.7	5.9	5.4	4.4	4.2	4.2	6.3
(15) Precipitation	PrecAvg	38.5	0.0	0.0	0.0	0.0	0.0	2.9	8.1	13.3	9.8	2.7	0.0	0.1
	PrecMax	64.6	0.4	0.4	0.0	0.0	0.4	10.6	15.6	23.6	31.5	11.8	0.3	1.9
	PrecMin	16.4	0.0	0.0	0.0	0.0	0.0	0.0	1.6	4.0	0.4	0.0	0.0	0.0
	PrecStd	11.6	0.1	0.1	0.0	0.0	0.1	2.6	3.9	5.0	5.3	2.3	0.1	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	101.9	104.2	103.6	99.0	94.9	91.8	90.0	91.7	95.2	97.0	97.4
		MCWB	67.4	67.4	68.5	70.5	73.6	76.9	78.8	79.8	81.4	79.9	74.0	70.5
	2%	DB	94.9	98.2	100.4	97.2	94.7	92.9	89.9	88.9	90.0	92.9	95.0	94.9
		MCWB	66.4	65.6	67.1	70.2	73.4	76.7	78.7	79.8	80.6	80.1	73.6	69.6
	5%	DB	92.2	95.2	96.7	93.3	91.4	91.2	88.7	87.8	89.2	91.3	93.1	92.8
		MCWB	65.0	64.6	66.4	70.2	73.4	76.6	78.7	79.7	80.4	80.0	73.7	68.7
	10%	DB	89.5	92.0	92.8	89.6	88.9	89.5	87.6	86.4	87.8	89.7	91.1	89.8
		MCWB	64.3	63.9	67.0	69.9	73.1	76.4	78.7	79.0	80.2	79.7	74.0	67.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	72.4	74.8	75.4	78.4	80.2	81.6	81.8	82.8	83.4	81.0	76.7
		MCDB	87.0	91.1	92.3	92.4	88.0	88.9	87.8	87.5	89.2	91.5	90.0	86.7
	2%	WB	69.7	69.8	72.3	73.4	76.9	79.2	80.1	80.8	81.8	82.0	79.6	74.7
		MCDB	85.2	85.6	88.4	89.1	87.0	87.2	86.7	86.7	87.9	89.4	87.9	83.4
	5%	WB	68.1	68.2	70.6	72.2	75.7	78.5	79.6	80.1	80.9	81.3	78.4	72.8
		MCDB	81.6	83.0	86.0	86.1	85.3	86.1	85.9	85.6	87.1	88.5	85.9	81.8
	10%	WB	66.8	67.0	69.2	71.0	74.6	77.8	79.0	79.4	80.2	80.6	77.1	71.3
		MCDB	80.4	81.1	83.8	83.6	83.8	85.3	84.9	84.6	85.9	87.4	84.1	81.5
(35) Mean Daily Temperature Range	5% DB	MDBR	26.4	27.3	26.4	23.1	19.5	15.2	12.5	12.0	13.6	15.3	21.4	25.1
		MCDBR	30.5	31.8	32.4	28.1	22.7	16.8	13.7	13.3	15.1	16.6	24.5	29.3
		MCWBR	12.5	11.0	9.8	8.4	7.0	5.5	5.7	6.2	7.4	7.1	10.4	12.1
	5% WB	MCDBR	27.4	27.4	27.4	24.1	19.2	15.0	12.8	12.5	14.2	15.4	20.3	25.9
		MCWBR	12.0	10.3	9.5	8.0	6.6	5.7	6.0	6.4	7.5	7.3	9.1	11.0
(40) Clear-Sky Solar Irradiance	taub		0.524	0.568	0.656	0.659	0.694	0.705	0.637	0.571	0.559	0.593	0.535	0.503
	taud		1.921	1.825	1.659	1.708	1.651	1.703	1.856	2.029	2.042	1.926	1.996	2.023
	Ebn at Noon		246	241	224	222	211	206	221	239	244	232	241	248
	Edh at Noon		60	68	82	78	81	76	65	56	55	61	55	53
(44) All-Sky Solar Radiation	RadAvg		1711	1914	2167	2272	2197	1970	1700	1623	1661	1776	1703	1623
	RadStd		88	91	70	59	74	65	83	83	86	80	81	78

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page